

Task 1: Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.10240]
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C:\Windows\system32>netsh
netsh>interface
In future versions of Windows, Microsoft might remove the Netsh functionality
for TCP/IP.

Microsoft recommends that you transition to Windows PowerShell if you currently
use netsh to configure and manage TCP/IP.

Type Get-Command -Module NetTCPIP at the Windows PowerShell prompt to view
a list of commands to manage TCP/IP.

Visit http://go.microsoft.com/fwlink/?LinkId=217627 for additional information
about PowerShell commands for TCP/IP.
netsh interface>ipv6 show interface

Idx      Met      MTU      State      Name
-----
1         50      4294967295 connected Loopback Pseudo-Interface 1
7         10       1500     connected Ethernet
4         50       1280     disconnected isatap.{FC359C31-C747-4837-9B7D-43FAA8EC1380}

netsh interface>ipv6 set address Ethernet 2001:db8::10

netsh interface>ipconfig /all
The following command was not found: ipconfig /all.
netsh interface>exit
```

```
Administrator: Command Prompt

C:\Windows\system32>ipconfig /all

Windows IP Configuration

Host Name . . . . . : win
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . . :
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-42-48-7C
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:db8::10(Preferred)
Link-local IPv6 Address . . . . . : fe80::c5f8:1105:dc85:58f6%7(Preferred)
IPv4 Address. . . . . : 192.168.56.102(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 50855075
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-D6-25-E4-08-00-27-42-48-7C
DNS Servers . . . . . : fec0:0:0:ffff::1%1
                       fec0:0:0:ffff::2%1
                       fec0:0:0:ffff::3%1
NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.{FC359C31-C747-4837-9B7D-43FAA8EC1380}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :
Description . . . . . : Microsoft ISATAP Adapter #2
Physical Address. . . . . : 00-00-00-00-00-00-E0
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

C:\Windows\system32>
```

Task 2: Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Hint: Use the ping6 or ping -6 command for IPv6 testing.

A unique local address always starts with fd (the fd00::/8 range)—it's IPv6's version of private addresses like 192.168.x.x.

Check existing IPv6 configuration on both PCs

On **PC1 (Host)**, run:

```
ipconfig /all
```

Ethernet adapter Ethernet 2 (VirtualBox Host-Only Ethernet Adapter):

Link-local IPv6 Address - : fe80::a9e7:ada9:7f48:16fe%13 (Preferred)

IPv4 Address - : 192.168.56.105 (Preferred)

```
Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . : 
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-0D
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a9e7:ada9:7f48:16fe%13(Preferred)
IPv4 Address. . . . . : 192.168.56.105(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 252313639
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-E8-2D-54-84-2A-FD-94-53-CD
NetBIOS over Tcpip. . . . . : Enabled
```

Need to change the IPV6 address in PC1 to match PC2

```

C:\Windows\System32>netsh
netsh>interface
netsh interface>ipv6 show interface

Idx      Met      MTU      State      Name
-----
1         75      4294967295 connected Loopback Pseudo-Interface 1
7         5        1500 disconnected Ethernet
3         55      1500 connected Wi-Fi
8         25      1500 disconnected Local Area Connection* 1
9         65      1500 disconnected Bluetooth Network Connection
4         25      1500 disconnected Local Area Connection* 2
13        25      1500 connected Ethernet 2

netsh interface>ipv6 set address "Ethernet 2" fd12:3456:789a:1::1

netsh interface>ipconfig /all
The following command was not found: ipconfig /all.
netsh interface>exit

```

On **PC2 (VM)**, ran:

```
netsh interface ipv6 set address "Ethernet" fd12:3456:789a:1::10
```

Result—Ethernet adapter Ethernet

IPv6 Address - : 2001:db8::10

IPv6 Address - : fd12:3456:789a:1::10

Link-local IPv6 Address - : fe80::c5f8:1105:dc85:58f6%7

```

Administrator: Command Prompt

C:\Windows\system32>netsh
netsh>interface
In future versions of Windows, Microsoft might remove the Netsh functionality
for TCP/IP.

Microsoft recommends that you transition to Windows PowerShell if you currently
use netsh to configure and manage TCP/IP.

Type Get-Command -Module NetTCPIP at the Windows PowerShell prompt to view
a list of commands to manage TCP/IP.

Visit http://go.microsoft.com/fwlink/?LinkId=217627 for additional information
about PowerShell commands for TCP/IP.
netsh interface>ipv6 set address Ethernet fd12:3456:789a:1::10

netsh interface>exit

C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2001:db8::10
    IPv6 Address. . . . . : fd12:3456:789a:1::10
    Link-local IPv6 Address . . . . . : fe80::c5f8:1105:dc85:58f6%7
    IPv4 Address. . . . . : 192.168.56.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Tunnel adapter isatap.{FC359C31-C747-4837-9B7D-43FAA8EC1380}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Windows\system32>

```

Verify the change on PC1

ipconfig /all

Confirmed new output under Ethernet 2:

IPv6 Address - fd12:3456:789a:1::1

Link-local IPv6 Address - : fe80::a9e7:ada9:7f48:16fe%13(Preferred)

```
Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-0D
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv6 Address. . . . . : fd12:3456:789a:1::1(Preferred)
    Link-local IPv6 Address . . . . . : fe80::a9e7:ada9:7f48:16fe%13(Preferred)
    IPv4 Address. . . . . : 192.168.56.105(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 252313639
    DHCPv6 Client DUID. . . . . : 00-01-01-00-31-E8-2D-54-84-2A-FD-94-53-CD
    NetBIOS over Tcpip. . . . . : Enabled
```

Both PCs now shared the same subnet prefix (fd12:3456:789a:1::/64), with unique host IDs (::1 on PC1, ::10 on PC2).

Ping test between PC1 and PC2

From **PC1 (Host)** → PC2:

ping -6 fd12:3456:789a:1::10

```
C:\Windows\System32>ping -6 fd12:3456:789a:1::10

Pinging fd12:3456:789a:1::10 with 32 bytes of data:
Reply from fd12:3456:789a:1::10: time=4ms
Reply from fd12:3456:789a:1::10: time=3ms
Reply from fd12:3456:789a:1::10: time=3ms
Reply from fd12:3456:789a:1::10: time=2ms

Ping statistics for fd12:3456:789a:1::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

From **PC2 (VM)** → PC1:

ping -6 fd12:3456:789a:1::1

```
C:\Windows\system32>ping -6 fd12:3456:789a:1::1

Pinging fd12:3456:789a:1::1 with 32 bytes of data:
Reply from fd12:3456:789a:1::1: time=2ms
Reply from fd12:3456:789a:1::1: time=3ms
Reply from fd12:3456:789a:1::1: time=1ms
Reply from fd12:3456:789a:1::1: time=1ms

Ping statistics for fd12:3456:789a:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Both pings returned successful replies, confirming Layer 3 connectivity between the host and the VM using unique local IPv6 addresses over the VirtualBox host-only network.

Task 3: Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Constraint: Document the steps and the observed address behavior in a text file.

Ran on the Wi-Fi adapter:

```
ipconfig /all
```

Result — Wireless LAN adapter Wi-Fi:

IPv6 Address - : 2405:201:200f:c1cf:8809:a17c:6516:6cf4(Preferred)

Temporary IPv6 Address - : 2405:201:200f:c1cf:538:e0e2:2cb1:91ef(Preferred)

Link-local IPv6 Address - : fe80::7bb7:f9b4:f160:a70e%3(Preferred)

Two separate global IPv6 addresses are visible on the same prefix (2405:201:200f:c1cf::/64):

- A regular auto-configured IPv6 Address, based on the interface identifier
- A Temporary IPv6 Address, randomly generated for privacy

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek RTL8821CE 802.11ac PCIe Adapter
Physical Address. . . . . : 5C-3A-45-48-0A-1B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2405:201:200f:c1cf:8809:a17c:6516:6cf4(Preferred)
Temporary IPv6 Address. . . . . : 2405:201:200f:c1cf:538:e0e2:2cb1:91ef(Preferred)
Link-local IPv6 Address . . . . . : fe80::7bb7:f9b4:f160:a70e%3(Preferred)
IPv4 Address. . . . . : 192.168.29.218(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, July 26, 2026 1:57:51 PM
Lease Expires . . . . . : Monday, July 27, 2026 4:41:18 PM
Default Gateway . . . . . : fe80::4e93:a6ff:fe8d:56c6%3
                          192.168.29.1
DHCP Server . . . . . : 192.168.29.1
DHCPv6 IAID . . . . . : 56375877
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-E8-2D-54-84-2A-FD-94-53-CD
DNS Servers . . . . . : 2405:201:200f:c1cf::c0a8:1d01
                          192.168.29.1
                          2405:201:200f:c1cf::c0a8:1d01
NetBIOS over Tcpip. . . . . : Enabled
```

PC restarted and noted change in the temporary IPv6 address

```
Physical Address. . . . . : 5C-3A-45-48-0A-1B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2405:201:200f:c1cf:8809:a17c:6516:6cf4(Preferred)
Temporary IPv6 Address. . . . . : 2405:201:200f:c1cf:8177:80a5:aac:4b28(Preferred)
Link-local IPv6 Address . . . . . : fe80::7bb7:f9b4:f160:a70e%3(Preferred)
IPv4 Address. . . . . : 192.168.29.218(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, July 26, 2026 8:16:58 PM
Lease Expires . . . . . : Monday, July 27, 2026 8:33:24 PM
Default Gateway . . . . . : fe80::4e93:a6ff:fe8d:56c6%3
                          192.168.29.1
DHCP Server . . . . . : 192.168.29.1
DHCPv6 IAID . . . . . : 56375877
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-E8-2D-54-84-2A-FD-94-53-CD
DNS Servers . . . . . : 2405:201:200f:c1cf::c0a8:1d01
                          192.168.29.1
                          2405:201:200f:c1cf::c0a8:1d01
NetBIOS over Tcpip. . . . . : Enabled
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 5C-3A-45-48-0A-1C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

	Before	After
Temporary IPv6 Address	2405:201:200f:c1cf:538:e0e2:2cb1:91ef	2405:201:200f:c1cf:8177:80a5:aac:4b28(new, changed)

Regular IPv6 Address	2405:201:200f:c1cf:8809:a17c:6516:6cf4	Same or regenerated based on interface identifier
Link-local Address	fe80::7bb7:f9b4:f160:a70e%3	Unchanged

Task 4: Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

```
C:\Users\srikantm05\Desktop>zomato_search.py
Enter restaurant name (or part of it): zo

Matching results:
Zomato Kitchen Vadodara -> IPv6: fd12:3456:789a:1::1

C:\Users\srikantm05\Desktop>zomato_search.py
Enter restaurant name (or part of it): mc

Matching results:
McDonald's Alkapuri -> IPv6: fd12:3456:789a:1::4

C:\Users\srikantm05\Desktop>_
```

Task 5: Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario (AI-Generated)

Network Prefix:
2001:db8:1234::/48

Requirement:
Divide the /48 network into at least three /64 subnets for different departments.

Subnet 1 – Administration

Network: 2001:db8:1234:1::/64

Example Host Address: 2001:db8:1234:1::10

Subnet 2 – Faculty

Network: 2001:db8:1234:2::/64

Example Host Address: 2001:db8:1234:2::20

Subnet 3 – Students

Network: 2001:db8:1234:3::/64

Example Host Address: 2001:db8:1234:3::30

Selected Subnet for Configuration:

Subnet 2 (Faculty) Assigned IPv6 Address: 2001:db8:1234:2::20/64

Manual Configuration Steps

Windows:

1. Open Command Prompt as Administrator.
2. Display available network interfaces:
`netsh interface ipv6 show interfaces`
3. Assign the IPv6 address:
`netsh interface ipv6 add address "Ethernet 2" 2001:db8:1234:3::30`
4. Verify the configuration:
`ipconfig /all`

Verification:

After running the verification command, the configured IPv6 address (2001:db8:5678:3::35/64) should appear under the selected network interface.

Note:

The IPv6 prefix 2001:db8::/32 is reserved for documentation and example networks. It is intended for educational purposes, lab exercises, and assignments and is not routable on the public Internet.

Network Prefix: 2001:db8:1234::/48

This is the "starting block"—the big chunk of address space we have to work with. The /48 means the first 48 bits are fixed (the network portion), leaving the remaining 80 bits (out of 128 total in IPv6) free to create subnets and assign hosts.

Requirement: Divide into at least three /64 subnets

2001 :	db8 :	1234 :	0002 :	0000 :	0000 :	0000 :	0000
network			subnet	host portion			

Going from /48 to /64 means you have 16 bits ($64 - 48 = 16$) to play with for creating individual subnets—that's 65,536 (2^{16}) possible /64 subnets from just this one /48 block. The scenario only uses 3 of them.

- Subnet 1 - Administration
 - Network: 2001:db8:1234:1::/64
 - Example Host Address: 2001:db8:1234:1::10
- Subnet 2 - Faculty
 - Network: 2001:db8:1234:2::/64
 - Example Host Address: 2001:db8:1234:2::20
- Subnet 3 - Students
 - Network: 2001:db8:1234:3::/64
 - Example Host Address: 2001:db8:1234:3::30

Selected Subnet for Configuration:

Subnet 2 (Faculty) Assigned IPv6 Address: 2001:db8:1234:2::20/64

```
Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix  . : 
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-0D
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:db8:1234:2::20(Preferred)
IPv6 Address. . . . . : fd12:3456:789a:1::1(Preferred)
Link-local IPv6 Address . . . . . : fe80::a9e7:ada9:7f48:16fe%13(Preferred)
IPv4 Address. . . . . : 192.168.56.105(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 252313639
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-E8-2D-54-84-2A-FD-94-53-CD
NetBIOS over Tcpip. . . . . : Enabled
```

